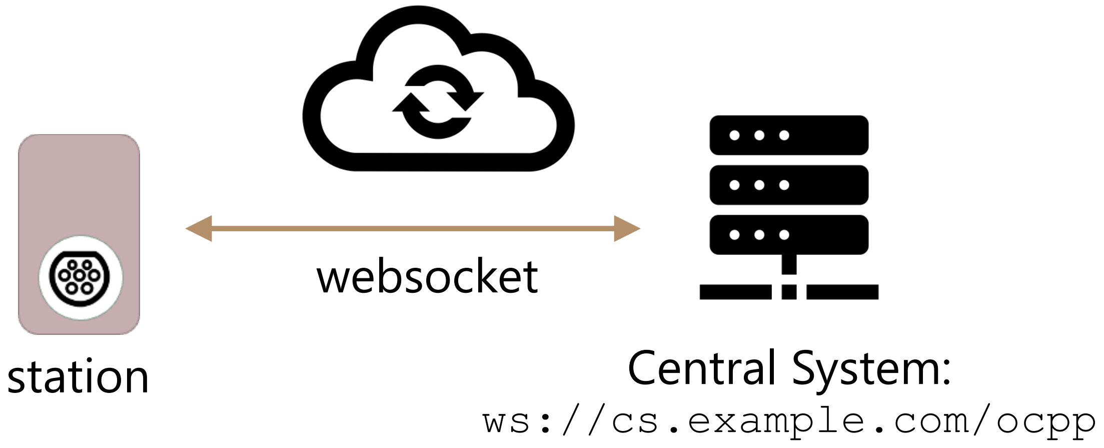
2. CONNECTING CHARGING STATIONS TO A CENTRAL SYSTEM

Connecting Stations to a Central System

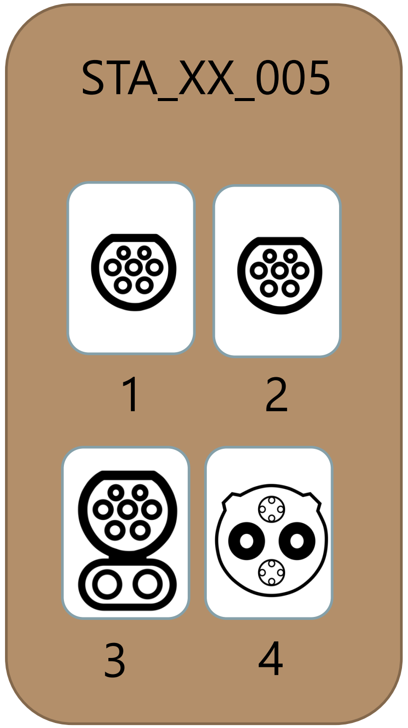


OCPP 1.6 supports two types of protocols for communication between a station and a central system: **SOAP** and **WebSockets**. In practice, the de facto standard used by most systems is WebSockets.

The Central System acts as a **WebSocket server** and the station acts as **a WebSocket client**.

1. The Central System has an URL such as `ws://cs.example.com/ocpp` for the OCPP server
2. This same URL is configured to the station, which opens a new WebSocket connection to the Central System
3. OCPP defines then what kind of JSON messages can be sent through the WebSocket connection

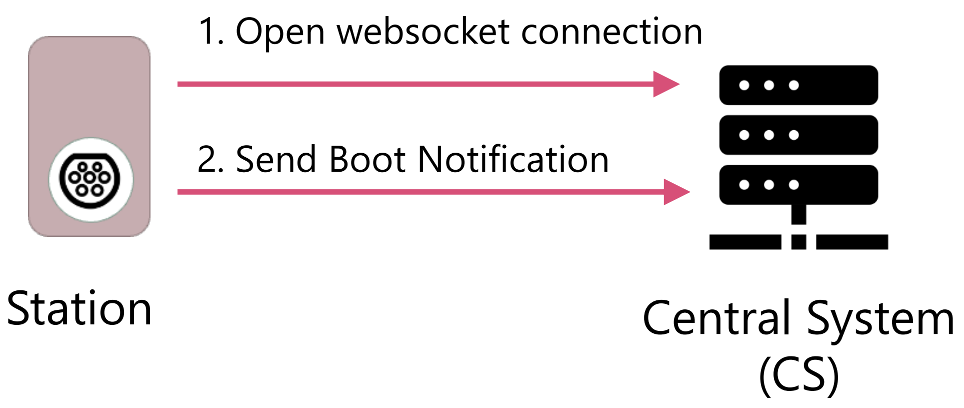
Identity and Connector IDs



In OCPP terminology, a charging station always has a unique identifier called an **"Identity"**. An example of an identity could be STA_XX_005.

Each connector of a station has a **Connector ID**, which is a number from 1 to N.

Connecting Stations to a Central System

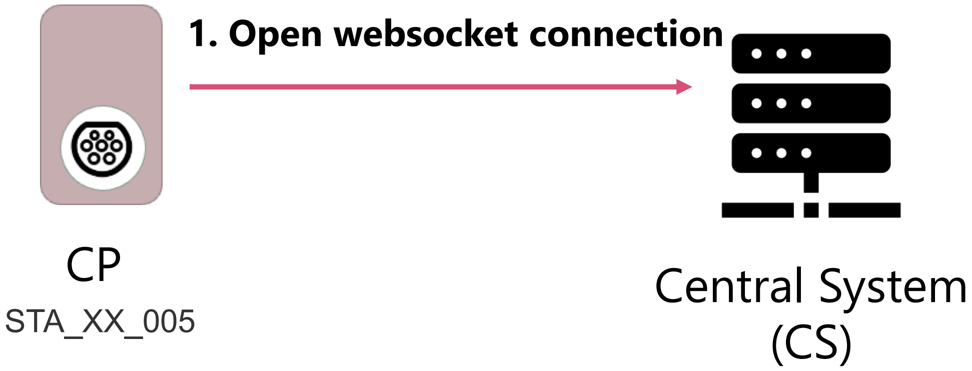


When a station **is powered on**, it should automatically **connect to the central system** in the following way::

1. The station checks its internal configuration for the URL of the central system and tries to open a new **WebSocket** connection there.

2. After the connection is successfully established, the station sends a '**Boot Notification**' message to the central system, providing details about the station that is now opening the connection.

WebSocket Connection

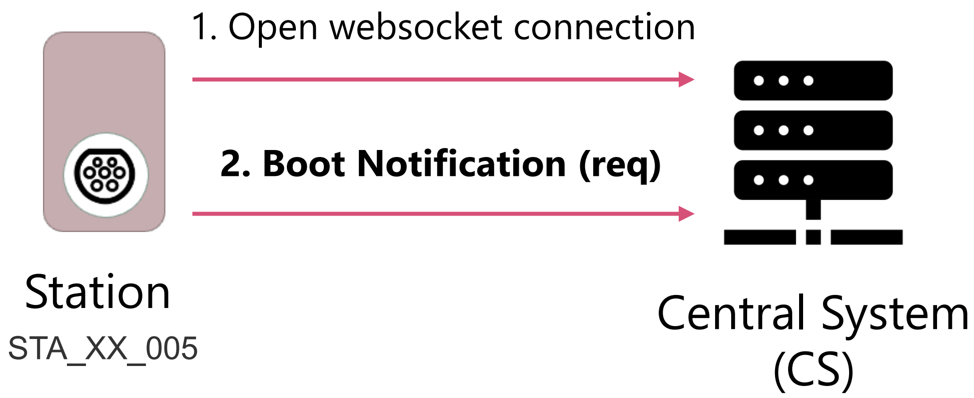


The Central System (CS) has an URL where the OCPP service is running, for example **ws://cs.example.com/ocpp**

The charging station has **a unique identity** that is sends as a part of the WebSocket (WS) connection URL it opens towards the CS.

For example, if s station's **identity** is **STA_XX_005**, it would open connection to the CS with the URL **ws://cs.example.com/ocpp/STA_XX_005**

Boot Notification Request



After the WebSocket connection has been established, the station sends a 'Boot Notification' request message to the central system.

[2, "19223201", "BootNotification", {"chargePointVendor": "VendorX", "chargePointModel": "SingleSocketCharger"}]

When the station opened the WebSocket connection to the CS, it included its identity in the URL. In the 'Boot Notification' message, the station provides more details, such as the **vendor** and **model** of the station, as well as additional information.

Structure of an OCPP message

CALL message format:

[<MessageTypeId>,

”<UniqueId>”, “<Action>”

{<Payload>}]

Example:

[2, "19223201", "BootNotification",

{"chargePointVendor": "VendorX",

"chargePointModel": ”V-MOD1-A"}]

As we saw in the previous example, a Boot Notification message sent by a CP has different parameters

[**2**, "19223201", "BootNotification", {"chargePointVendor": "VendorX", "chargePointModel": " V-MOD1-A"}]

First parameter **2** is a message type. Possible **message types** are:

2	CALL	Client-to-Server
3	CALLRESULT	Server-to-Client
4	CALLERROR	Server-to-Client

Since our message was sent by the CP (client) to the CS (server), the message type was **2 (CALL)**

Structure of an OCPP CALL message

CALL message format:

[<MessageTypeId>,

”<UniqueId>”, “<Action>”

{<Payload>}]

Example:

[2, "19223201", "BootNotification",

{"chargePointVendor": "VendorX",

"chargePointModel": ”V-MOD1-A"}]

The second parameter **"19223201”** is a message ID that binds together CALL and CALLRESULT

[2, **"19223201"**, "BootNotification", {"chargePointVendor": "VendorX", "chargePointModel": " V-MOD1-A"}]

When a station sends a message (CALL) to the Central System with ID **"19223201”,** the Central System must then include this same Message ID back to the CALLRESULT.

This way both the client and the server always know, which **response** (CALLRESULT) is linked to which **request** (CALL)

Structure of an OCPP CALL message

CALL message format:

[<MessageTypeId>,

”<UniqueId>”, “<Action>”

{<Payload>}]

Example:

[2, "19223201", "BootNotification",

{"chargePointVendor": "VendorX",

"chargePointModel": ”V-MOD1-A"}]

The third parameter **"BootNotification”** tells which kind of message we are sending.

[2, "19223201", **"BootNotification"**, {"chargePointVendor": "VendorX", "chargePointModel": " V-MOD1-A"}]

When the Central System receives a Boot Notification message from a Station, it knows that this Station has now been rebooted. It can then perform different actions needed by the Central System, for example mark that the Station is now online.

Structure of an OCPP CALL message

CALL message format:

[<MessageTypeId>,

”<UniqueId>”, “<Action>”

{<Payload>}]

Example:

[2, "19223201", "BootNotification",

{"chargePointVendor": "VendorX",

"chargePointModel": ”V-MOD1-A"}]

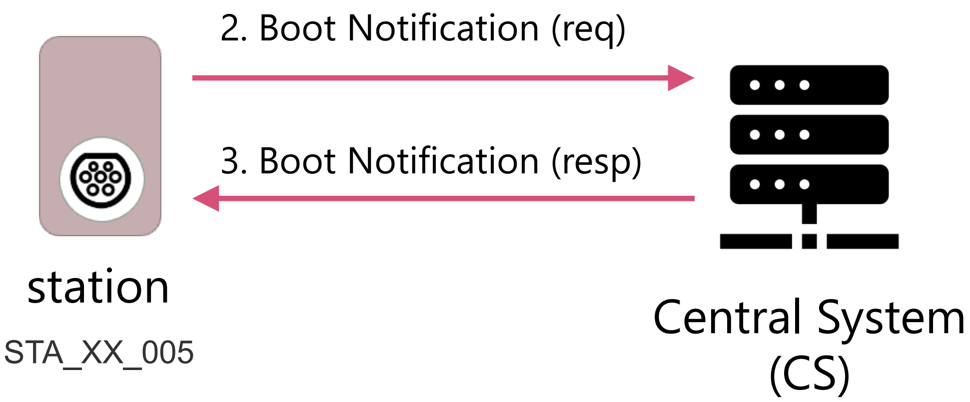
The last in the message is then the actual payload of the BootNotifcation message.

[2, "19223201", "BootNotification", **{"chargePointVendor": "VendorX", "chargePointModel": " V-MOD1-A"}**]

In this example, the payload has two mandatory parameters: vendor, which tells which company has made the charging station, and model, which tells the detailed model number of this station.

The station can also send optional parameters in the message, such as serial number, firmware version, ICCID and IMSI of a SIM card, and energy meter serial and type.

Boot Notification Response



When the Central System receives a Boot Notification request from a station, it may **Accept** or **Reject** the request. Many Central Systems, for example have a security feature where they allow only CPs with a whitelisted identity to connect to the CS.

[3, "19223201", {"status": "Accepted", "interval": 60, "currentTime": "2022-12-05T12:31:14Z"}]

The Boot Notification response also allows the Central System to tell the stations which time they should use and an interval (in seconds) on how often the CP should report heartbeats and meter values to the CS.

All Boot Notification request fields

Field	Card.	Description
chargePointVendor	1..1	Required. Name of the vendor who has manufactured the charging stations. Usually a company name such as *Charger Inc.*
chargePointModel	1..1	Required. Model name or number of a charging station. Usually a technical model identifier such as *singe_acbox_22*
chargeBoxSerialNumber	0..1	Optional. Deprecated field from previous OCPP versions
chargePointSerialNumber	0..1	Optional. Serial number of the charging station
firmwareVersion	0..1	Optional. Firmware version installed to the station
iccid	0..1	Optional. ICCID of the SIM-card in station's modem
imsi	0..1	Optional. IMSI of the SIM-card in station's modem
meterSerialNumber	0..1	Optional. Serial number of the energy meter of the station
meterType	0..1	Optional. Type of the energy meter of the station